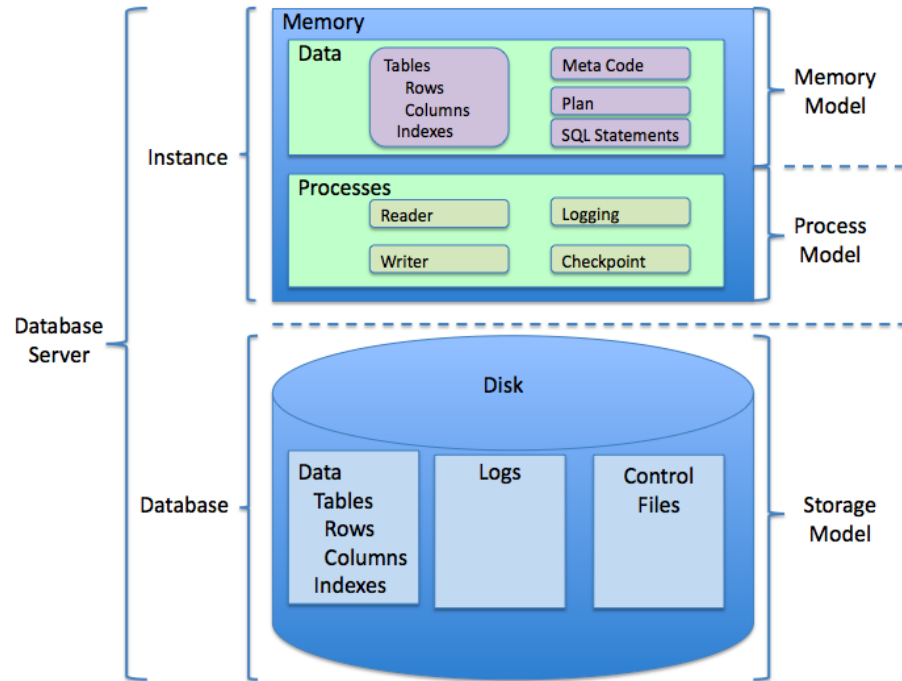


Database Management Systems



By: Assistant Professor Malik M Ali

Course Assessment Policy

- Theory Part

- Assignments (3) 20
- Quiz (4) 10

For assignments:

Timely submission is absolutely necessary

In case of late submission submit your assignment at kcclad08@gmail.com

- Midterm and Final Exam20, 50

Course Assessment Policy

- Lab Part

- Lab Performance52
- Mid Exam12
- Final Exam06
- Lab Project.....30

Course Website Information

1. LMS
2. YouTube Channel Name: Malik Ali Academy
3. Channel Link

<https://www.youtube.com/channel/UC-Rh64DwoESImOXBEUKQcrg/playlists>

Plagiarism

- Submitting copied assignment / project is a serious crime and will be dealt seriously. You will be awarded negative marks for such an attempt
- You are supposed to provide acceptable references for the ideas discussed in assignments

Importance of Time

- You have to come in the class with in the first 10 minutes in order to get marked as present
- 75% presence is mandatory
- All assignments, quizzes, presentations and project submission should be on time.
- Counseling hours: to be discussed

Books

1. Fundamentals of Database Systems 7th Ed by Ramez Elmasri
2. Database Systems – A practical approach to Design, Implementation and Management By Thomas M Connolly
3. Beginning SQL Queries From Novice to Professional by APress Publishers

Course Objectives

They key objectives of this course include:

- To understand the different issues involved in the design and implementation of a database system.
- To study the physical and logical database designs, database modeling, relational, hierarchical, and network models.
- To understand and use data manipulation language to query, update, and manage a database.
- To develop an understanding of essential DBMS concepts such as: database security, integrity, and concurrency.
- To design and build a simple database system and demonstrate competence with the fundamental tasks involved with modeling, designing, and implementing a DBMS.

Course Learning Outcomes

After successful completion of this course, the students should be able to:	PLO	BT Level
1. Explain the fundamental concepts of databases	2	C2
2. Apply different database models to design conceptual, logical, and physical databases	4	C3
3. Apply queries to extract information from databases	5	C3
4. Analyze user requirements to design a database for the given scenarios	4	C4

II. Understanding	III. Applying	IV. Analyzing
Demonstrate understanding of facts and ideas by organizing, comparing, translating, interpreting, giving descriptions, and stating main ideas.	Solve problems to new situations by applying acquired knowledge, facts, techniques and rules in a different way.	Examine and break information into parts by identifying motives or causes. Make inferences and find evidence to support generalizations.
• Classify • Compare • Contrast • Demonstrate • Explain • Extend • Illustrate • Infer • Interpret • Outline • Relate • Rephrase • Show • Summarize • Translate	• Apply • Build • Choose • Construct • Develop • Experiment with • Identify • Interview • Make use of • Model • Organize • Plan • Select • Solve • Utilize	• Analyze • Assume • Categorize • Classify • Compare • Conclusion • Contrast • Discover • Dissect • Distinguish • Divide • Examine • Function • Inference • Inspect • List • Motive • Relationships • Simplify • Survey

Course Outline

- Basic Terms and Concepts
- File system versus DBMS
- Architecture of a DBMS
- Data Models
- ER Models
- Types of Keys
- Functional dependencies
- Multivalued Dependencies
- Normalization
- Types of Joins
- Indexing and Constraints
- Transactions (Theory)
- Views and Authorizations
- OLAP
- Relational Algebra
- SQL using SQL Server 19
 - Data query
 - Data manipulation
 - Data definition
 - Data access control
 - Transactions
 - T-SQL

Data versus Information

Data

- Raw facts
 - Have not yet been processed to reveal their meaning to the end user
 - e.g. numbers, characters, graphics, images, audio, video
- Building blocks of information
- **Data management**
 - Generation, storage, and retrieval of data
- Data is what you store in *database*.

Information

- Produced by processing raw data to reveal its meaning
- Requires context
 - e.g. Total sale for month June, Total students at Bahria Karachi campus
- Bedrock of **knowledge**
- Should be accurate, relevant, and timely to enable good decision making
- Information is what you retrieve from a database.

Data

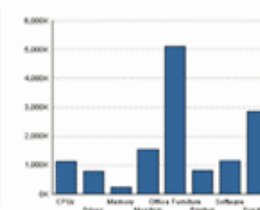
ID	Name	Address	City	State
1	Andrew	SW 16 th Ave	Orlando	FL
2	Robert	23 rd Terrace	Boston	MA
3	Bob	52nd Avenue	Bronx	NY
4	Lee Hung	52nd Avenue	Albany	NY
5	Richard	999 Palm Bay	Detroit	MI
6	Tony	15th Avenue	Austin	TX

Information – Pie Chart, Bar Graph Table

Total % Sales per Salesperson



Sales by Product Category



Total Sales - Salesperson

Salesperson	Total	Commission
Allan McEwen	\$59,872.40	\$20,955.34
Christopher Olsen	\$10,794.60	\$2,158.92
Danielle Greene	\$68,501.00	\$10,275.15
Janette King	\$23,034.60	\$8,062.11
Lindsey Smith	\$1,233.00	\$369.90
Louise Doran	\$384.00	\$115.20
Nanette Cambrault	\$42,283.20	\$8,456.64
Oliver Tuvault	\$46,257.00	\$6,938.55
Sarah Sewall	\$17,848.20	\$4,462.05
Total:	\$13,617,206	\$3,479,484

Types of Data

- **Unstructured data:** Data having no structure. For Example Websites, Videos
- **Semi-Structured data:** Data having some structure. For Example emails
- **Structured data:** Data having a structure. It can be shown in tabular format.
- For Example: Tables in a relational database

Definitions

Malik M Ali

- Database:

- Database is a shared collection of logically related data that is stored to meet the requirements of different users of an organization. Stored data is of two types.

- End user data: Raw facts

- Meta data: Data about data

- Database management system:

- is a software designed to manage database structure and controls access to the data stored in the database. For example: Oracle, SQL Server, DB2, MySQL, MS Access etc.

The Database System Environment

- The database system composed of 5 major parts.
 - **Hardware:** includes all of the system's physical devices, including computers, storage, printers, network devices and other devices like automated teller machines, ID readers and so on.
 - **Software:** three types of software are required
 - Operating system (Windows, Linux, Mac etc)
 - DBMS (Oracle, SQL Server, MySQL, DB2 etc)
 - Application programs: programs written by application developers to access and manipulate data. These programs can generate reports for decision making.
-

The Database System Environment

- ❑ **People:** it includes all users of the database system.
 - **System Administrator:** oversee the database system's general operations
 - **Database Administrator:** Manage the DBMS and ensure that the database is functioning properly. The DBA is responsible for
 - ❑ authorizing access to the database
 - ❑ Coordinating and monitoring its use
 - ❑ Acquiring software and hardware resources as needed.
 - ❑ Breach of security or poor system response time.
-

The Database System Environment

- **Database Designer:** Design the database structure. Designers' responsibilities include
 - Identification of the data to be stored in the database
 - Choosing appropriate structures to represent and store this data.
 - **Analyst Programmers:** Design and implement application programs to provide different functions of the database to the End-Users
 - **End Users:** People who use the application programs to run the organization's daily operations.
-

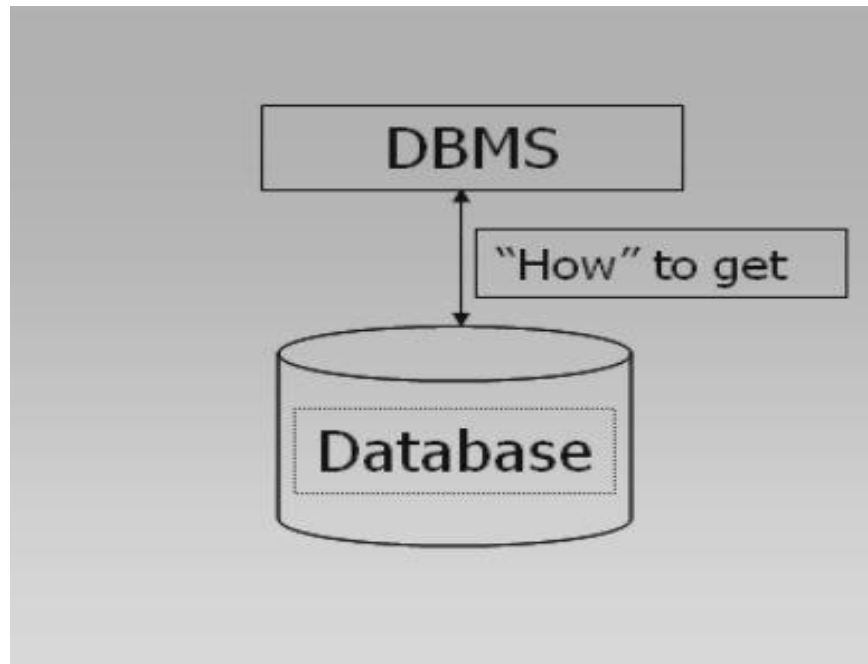
The Database System Environment

Malik M Ali

- ❑ **Procedures:** Rules that govern the design and use of the database system.
 - ❑ **Data:** collection of Facts stored in the database. Determining what data to enter into the database and how to organize those data is a vital part of the database designer's job.
-

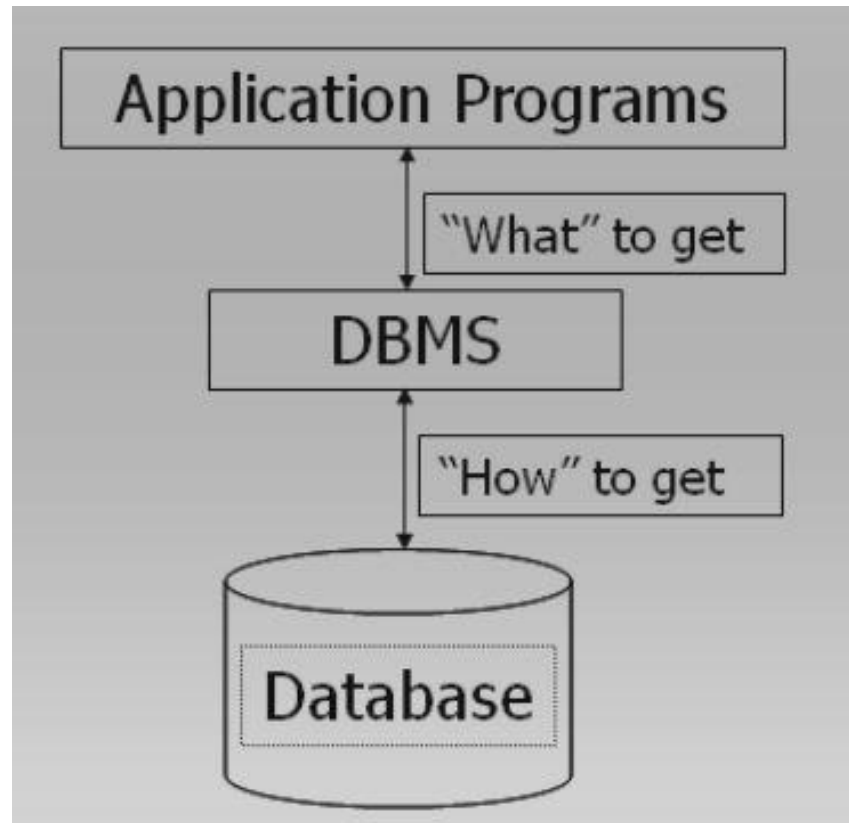
Interaction of Database System Components

- Different typical components of a database environment are shown in the figures below; they describe graphically the role of different types of users
- Database is used to store data and DBMS uses mechanisms to get data from the database



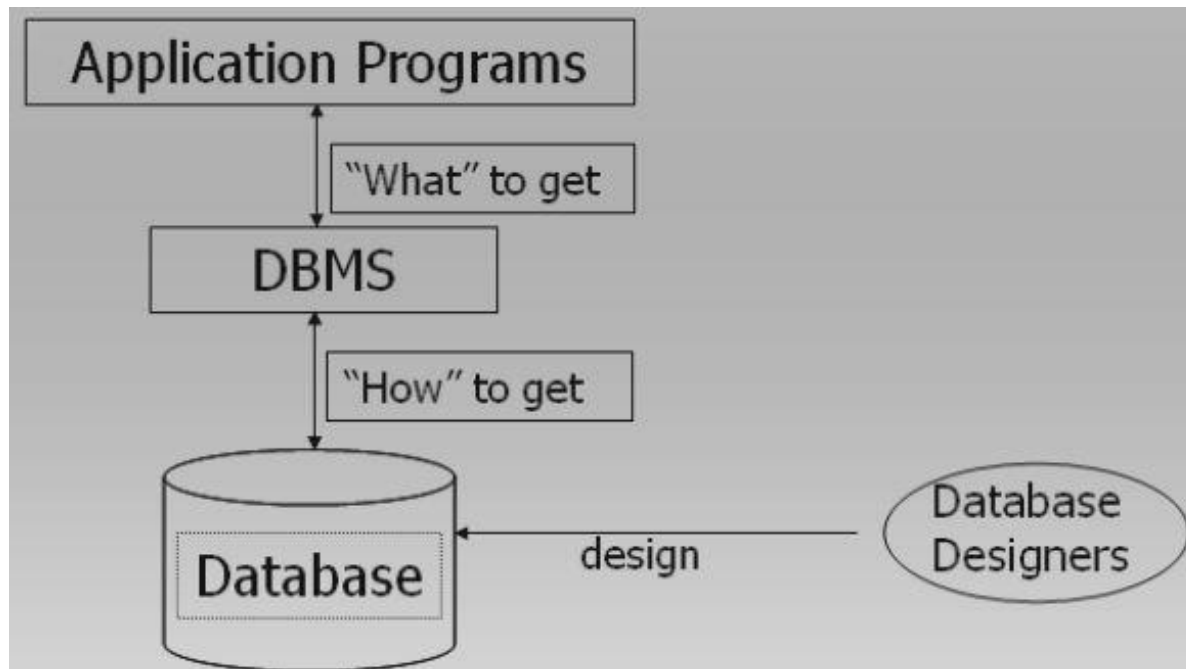
Interaction of Database System Components

- Application programs talk to the DBMS and ask for the data required



Interaction of Database System Components

- Database designers design and implement the database on the selected DBMS

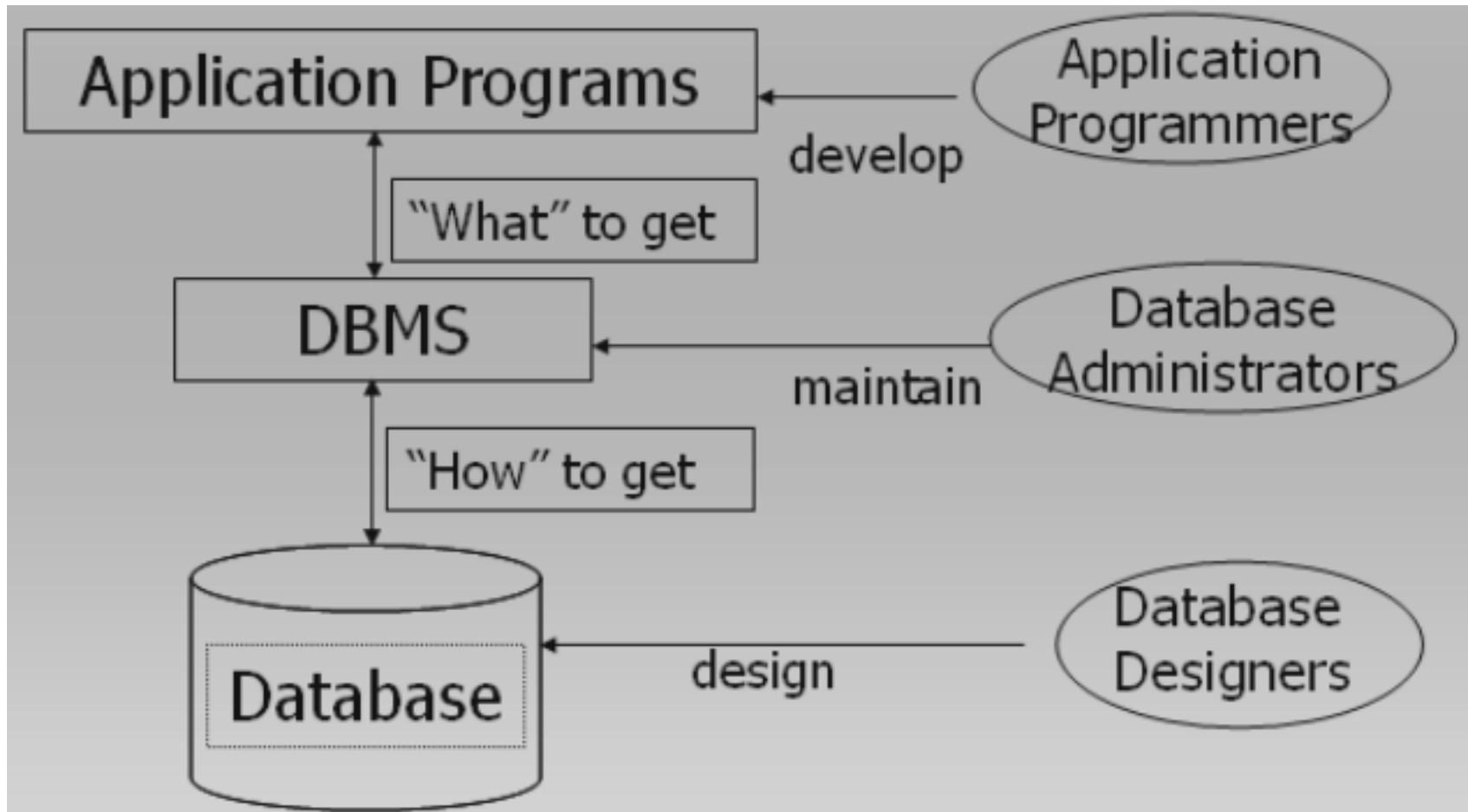


Interaction of Database System Components

Malik M Ali

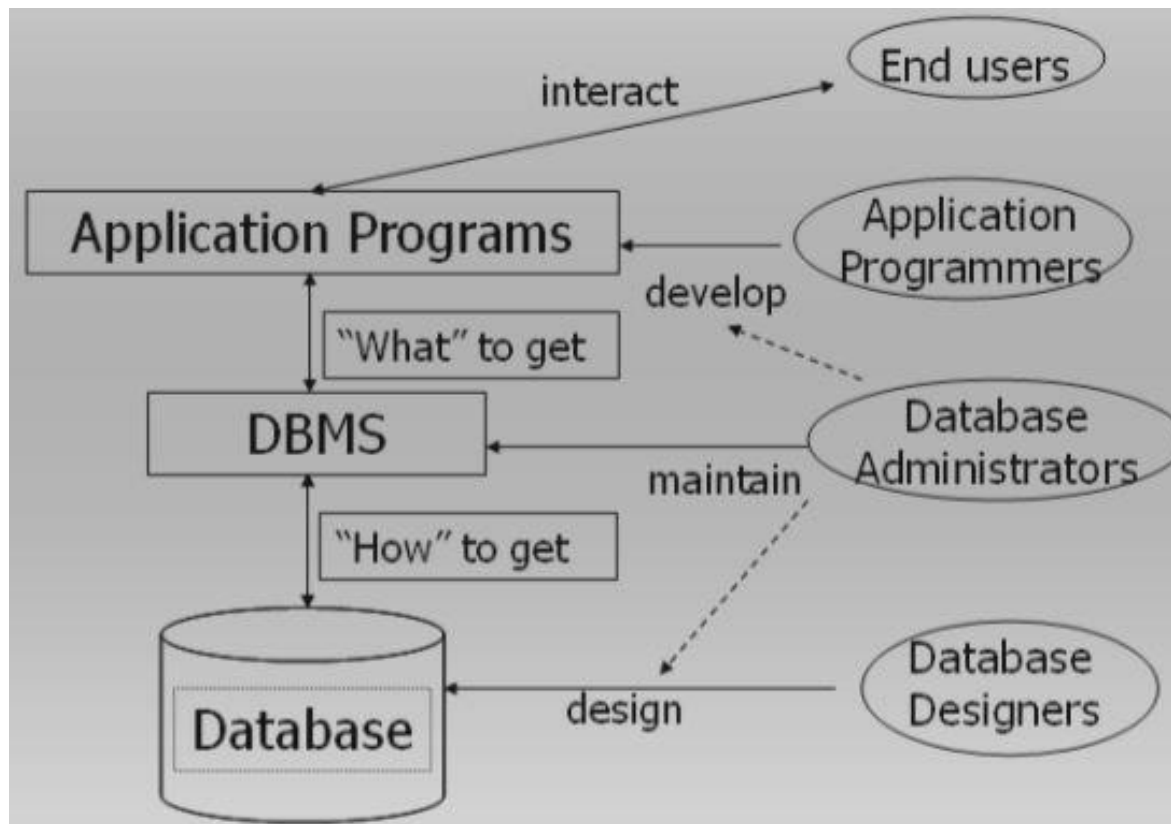
- Once Database has been implemented and is functioning properly in a production environment of an organization the Database Administrator takes over the charge and performs specific DBA related activities including:
 - ❑ Database maintenance
 - ❑ Database Backup
 - ❑ Grant and Revoke rights to database users
 - ❑ Monitoring of Running Jobs
 - ❑ Managing Print jobs
 - ❑ Ensuring quality of Service to all users
-

Interaction of Database System Components



Interaction of Database System Components

- DBA's interaction with other users



Interaction of Database System Components

- Database administrator can interact with the database designer during database design phase so that he has a clear idea of the database structure for easy reference in future.
 - This helps DBA perform different tasks related to the database structure.
 - DBA also interacts with the application programmers during the application development process and provides his services for better design of applications.
 - End users also interact with the system using application programs and other tools as specified in the description above
-

Responsibilities of a DBMS

Malik M Ali

The DBMS provides all the basic services required to organize and maintain the database in correct format. It includes management of

- **Data storage:** DBMS creates and manages the complex structures required for data storage. It is a job of DBMS to Move data to and from the physical data files as needed in an effective and efficient manner.
-

Responsibilities of a DBMS

Malik M Ali

- **Data security:** DBMS creates a security system that enforces user security and data privacy. Security rules determine which users can access the database, which data items each user can access and which data operations (read, insert, update, delete) the user can perform.
 - **Concurrent access:** Managing concurrent data access by multiple users, including provisions to prevent conflicting simultaneous updates
-

Responsibilities of a DBMS

Malik M Ali

- **Backup and recovery:** DBMS provide special utilities that allow the DBA to perform backup and restore procedures. Recovery features deals with recovering a database after a failure such as bad sector in the disk or a power failure
 - **Data integrity:** DBMS minimize data redundancy and maximize data consistency through integrity rules. Data relationships stored in the data dictionary and rules for Transaction management are used to enforce data integrity and consistency.
-

Responsibilities of a DBMS

Malik M Ali

- **Database access languages and APIs:** DBMS provides data access through a query language known as SQL (structured query language). It also provides application programming interfaces to high level languages like C, C++, Java, C#, VB.Net, Python etc.
 - **Database communication interfaces:** DBMS accepts end-user requests via multiple sources. For example, the DBMS might provide access to the database via the internet through the use of web browser
-